Data Analysis Report on Student Performance Dataset

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Abstract

This report analyzes a student performance dataset containing 1,000 records with features such as study hours and grades. We employed descriptive statistics, correlation analysis, and a linear regression model. Key findings include a correlation coefficient of 0.76 between study hours and grades, and a regression model with an R^2 value of 0.58.

1. Introduction

The academic performance of students is influenced by various factors, including study habits, attendance, and socio-economic background. Understanding these factors can help educators develop effective strategies to enhance student performance. This analysis focuses on a dataset sourced from a local educational institution, comprising 1,000 students with features such as study hours, grades, and demographic information. The primary objectives are to explore the relationship between study hours and grades, perform descriptive statistics, and build a regression model to predict grades based on study hours.

2. Exploratory Data Analysis

Descriptive statistics were computed for the dataset, revealing the following key metrics: - Mean study hours: 5.4 hours - Mean grades: 75.2 - Standard deviation of study hours: 2.1 - Standard deviation of grades: 10.5

Statistic	Study Hours	Grades
Mean	5.4	75.2
Standard Deviation	2.1	10.5
Minimum	1	50
Maximum	10	100

Table 1: Descriptive statistics of study hours and grades.

The distribution of study hours is approximately normal, while grades exhibit a slight right skew. A scatter plot indicates a positive correlation between study hours and grades.

3. Methods

To quantify the relationship between study hours and grades, we calculated the Pearson correlation coefficient:

$$r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2}\sqrt{\sum(Y_i - \bar{Y})^2}}$$

where X represents study hours and Y represents grades. The resulting correlation coefficient was $r = 0.76$, indicating a strong positive correlation.

We then fitted a linear regression model:

$$\text{Grades} = \beta_0 + \beta_1 \cdot \text{Study Hours} + \epsilon$$

where β_0 is the intercept, β_1 is the slope, and ϵ is the error term.

```python
import pandas as pd
import statsmodels.api as sm

data = pd.read_csv('student_performance.csv')
X = data['Study_Hours']
y = data['Grades']
X = sm.add_constant(X)  # Adds a constant term to the predictor
model = sm.OLS(y, X).fit()
print(model.summary())
```

Listing 1: Linear regression model fitting in Python

4. Results

The regression model yielded the following performance metrics: - R^2: 0.58 - Adjusted R^2: 0.57 - p-value for study hours: 0.0001

Metric	Value
R^2	0.58
Adjusted R^2	0.57
p-value (Study Hours)	0.0001

Table 2: Performance metrics of the regression model.

The model indicates that for every additional hour studied, grades increase by approximately 8.5 points.

5. Discussion

The analysis demonstrates a significant positive relationship between study hours and academic performance. The R^2 value suggests that the model explains 58% of the variance in grades, indicating a moderate fit. Limitations include potential confounding variables not accounted for, such as teaching quality and student motivation. Future work could explore these factors to enhance model accuracy.

6. Conclusion

In conclusion, this analysis highlights the importance of study hours in predicting student performance. The findings suggest that increasing study time can lead to improved grades, emphasizing the need for effective study strategies. Future research should consider additional variables to further refine the predictive model.

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